

Program 5

CI Integration by running Jenkins in a Docker container

Steps

1. Make sure docker compose is installed

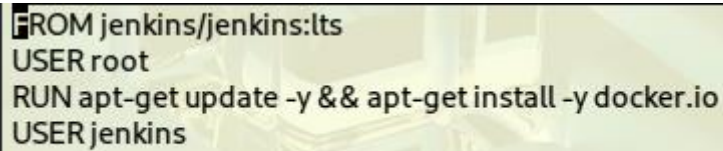
docker --version

docker compose version



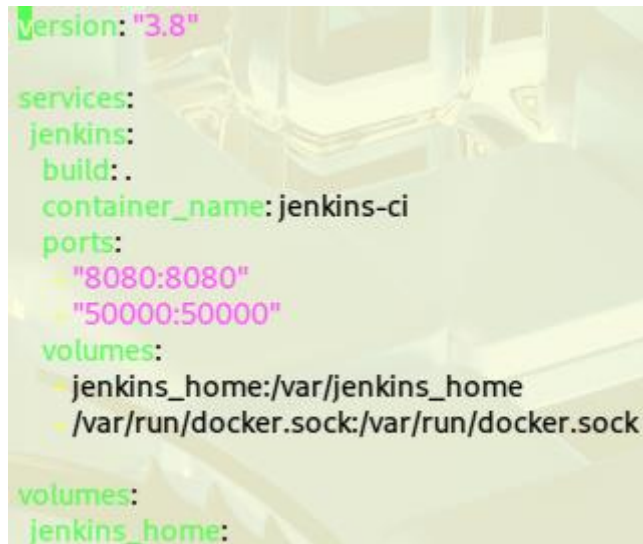
```
(1rv24mc045_ryan@ryan-kali) - [~/Documents/program6]
$ docker compose version
Docker Compose version v2.40.0
```

2. Create a dockerfile



```
FROM jenkins/jenkins:lts
USER root
RUN apt-get update -y && apt-get install -y docker.io
USER jenkins
```

3. Create a docker-compose.yml



```
Version: "3.8"

services:
  jenkins:
    build: .
    container_name: jenkins-ci
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock

volumes:
  jenkins_home:
```

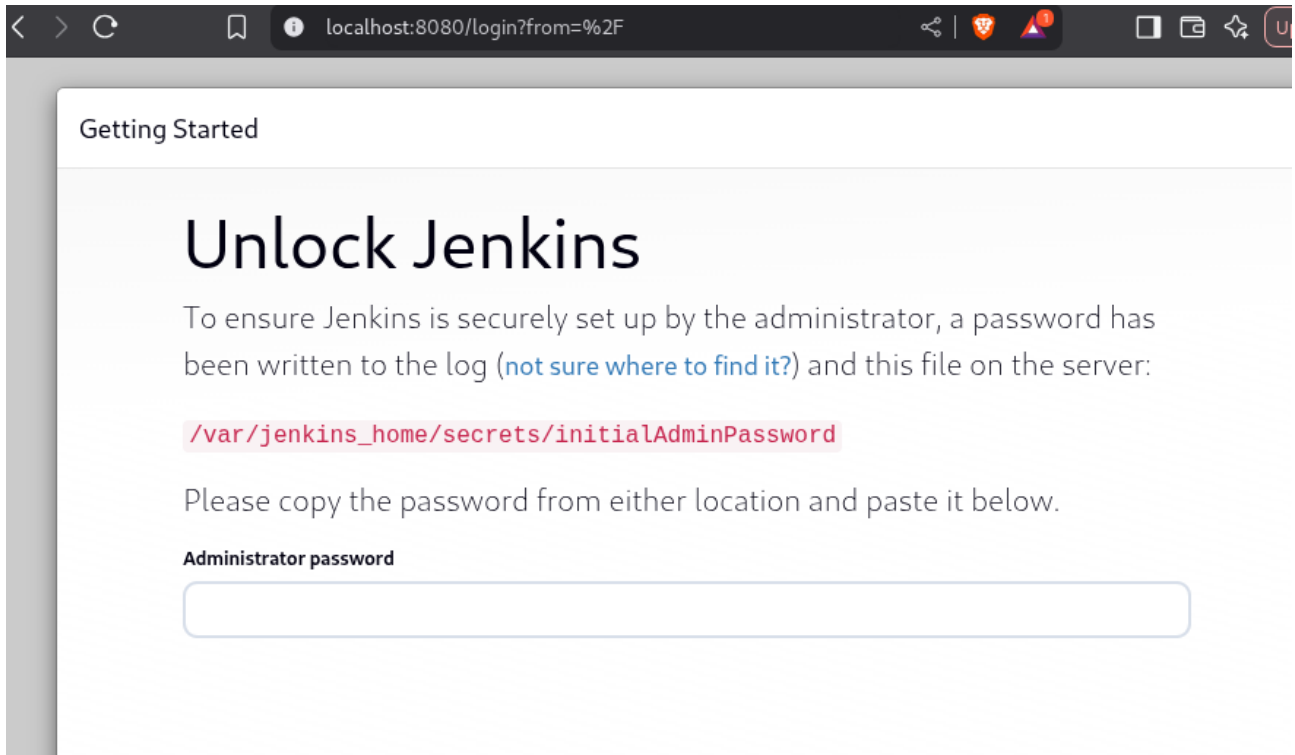
4. Build the jenkins container using the cmpose file

docker compose up -d

5. Navigate
the

```
(1rv24mc045_ryan@ ryan-kali) ~/Documents/program5
$ docker compose up -d
WARN[0000] /home/1rv24mc045_ryan/Documents/program5/docker-compose.yml: the attribute `version` is obsolete, it will
be ignored, please remove it to avoid potential confusion
[+] Building 0.0s (0/0)
[+] Running 4/4
✓ program5-jenkins      Built      0.0s
✓ Network program5_default Created    0.1s
✓ Volume program5_jenkins_home Created    0.0s
✓ Container jenkins-ci   Started   0.4s
```

<http://localhost:8080> to see the jenkins client



6. Delete the container and images afterwards

```
docker image rm program5-jenkins
docker container rm <id>
```

(1rv24mc045_ryan@ ryan-kali) ~/Documents/program5

\$ docker ps

CONTAINER ID	IMAGE	COMMAND NAMES	CREATED	STATUS	PORTS
e17d917008b5	program5-jenkins	"/usr/bin/tini -- /u..."	10 minutes ago	Up 10 minutes	0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp jenkins-ci

(1rv24mc045_ryan@ ryan-kali) ~/Documents/program5

\$ docker container stop e17

e17

(1rv24mc045_ryan@ ryan-kali) ~/Documents/program5

\$ docker container rm e17

e17

(1rv24mc045_ryan@ ryan-kali) ~/Documents/program5

\$ docker image rm program5-jenkins

Untagged: program5-jenkins:latest

Deleted: sha256:13725a30d1d43bd03dc8d0485a3734d186ed6877643152446b1f0224117a4c17